SOFTWARE REQUIREMENTS SPECIFICATION

RealmForge

Team 20
Aeolus Software

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Revision 0: October 14, 2026

Revision History

Name	Date	Reason For Changes	Version
Andy Huynh	2026-10-05	First Draft	1.0
Andreas Sideris	2026-10-10	P3 and P4 drafts	1.0
Andreas Sideris	2026-10-12	E3 draft	1.0
Andreas Sideris	2026-10-13	E2 draft	1.0

This document follows the requirements documentation structure presented in the Handbook of requirements and business analysis, by Bertrand Meyer.

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Goals

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Comment: *Goals are "needs of the target organization, which the system will address". While the development team is the principal user of the other books, the Goals book addresses a wider audience: essentially, all stakeholders.*

G.1 Context and overall objective

Comment: *High-level view of the project: organizational context and reason for building a system.*
Comment: *This chapter should not be empty!*

Goal 1. This is a goal example. If you need explicit (and automatic) numbering, you can use the definitions in the `.tex` template. Is is refined by 3

Requirement 2. *This is a requirement example. It illustrates how numbering is continuous and cross-types (if this is what you need).*

G.2 Current situation

Comment: *Current state of processes to be addressed by the project and the resulting system.*

Requirement 3. *This is a requirement example. It refines 1*

G.3 Expected benefits

Comment: *New processes, or improvement to existing processes, made possible by the projectâ€™s results.* Comment: *This chapter should not be empty!*

G.4 Functionality overview

Comment: *Overview of the functions (behavior) of the system. Principal properties only (details are in the System book).*

Setting properties: The system application starts up and the the user is shown options that will allow the user to pick and change various settings during the start of the application, so that the user can experiment with different variants of the game. Comment: *Less detail on the button pressing*

Traversal: After the player has changed some properties or just simply pressed the play option menu button, the game begins and the system displays the HUD as well other UI components such as the main game screen and travel options. The player can navigate to another area by pressing the up, down, left, or right menu option buttons go to another area indicated by the change in the background scenery.

Comment: *Branching patchs, endpoints, you need to win*

Combat: The player enters combat randomly when traversing through areas. Once in combat, the battle screen is shown as well as the enemy the player has to defeat. The player is shown options to attack or escape during the encounter. If the player chooses to attack the enemy, an attack action is displayed followed by the enemies opposing attack. If the player defeats the enemy, the player is rewarded with resources at the end. Comment: *You win or die*

Returning to the Hub: If the player escapes, then there is no reward and the player goes back to the main game screen. The gameplay loop of traveling and encounter enemies continues until the player returns to the starting zone (the main hub), with the return option or when the player is defeated. If the player returns to the starting zone, the player is allowed to spend the resources accumulated throughout the journey to construct buildings and upgrades that will enhance the experience of the next gameplay loop. Comment: *Change to only comming back when you die* Comment: *Immediately enter combat* Comment: *Remove randomization* Comment: *You know what your getting into, but not complete knowledge* Comment: *Higher risk, higher reward*

End of game: Once the player finishes spending resources, the player may return to the main gameplay loop of traveling and fighting enemies to gain resources. If the player is defeated, the game ends and player is returned to the start of the game where the player chooses options for changing parameters.

G.5 High-level usage scenarios

Comment: *Fundamental usage paths through the system.*

G.6 Limitations and exclusions

Comment: *Aspects that the system need not address.*

G.7 Stakeholders and requirements sources

Comment: *Groups of people who can affect the project or be affected by it, and other places to consider for information about the project and system.* **Comment:** *This chapter should not be empty!*

Environment

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Comment: *The Environment book describes the application domain and external context, physical or virtual (or a mix), in which the system will operate.*

E.1 Glossary

Comment: *Clear and precise definitions of all the vocabulary specific to the application domain, including technical terms, words from ordinary language used in a special meaning, and acronyms. This chapter should not be empty!*

- **User** - The main stakeholder that is operating the system; providing inputs and receiving outputs.
- **Player** - Is another name refering the user.
- **User-Interface (UI)** - The entire set of displayed service elements the player used to navigate the system.
- **Heads-Up-Display (HUD)** - The set of interface elements displayed ontop of the actually gameplay that contains vital information helpful to the player or user.
- **Central Processing Unit (CPU)** - The central processor for a computer.
- **Streaming Single Instruction, Multiple Data Extentions 4 (SSE4.2)** - A processor instruction set used by most modern processors.
- **Advanced Micro Devices (AMD)** - An American semiconductor company that produces processors.
- **Graphics Processing Unit (GPU)** - A computer's processor for displayed graphics.

- **Open Graphics Library (OpenGL)** - An application programming interface used to render computer graphics.
- **NVIDIA Corporation** - An American technology company that produces graphics processors.
- **Graphics Core Next (GCN)** - A series of graphics processor architectures that is used in AMD graphics processors.
- **Random Access Memory (RAM)** - Short-term memory used by a computer's processor.
- **Frames per Second (FPS)** - The number of graphical frames displayed by a computer program per second.
- **Personal Computer (PC)** - A computer owned by individuals, for personal use.
- **Verification and Validation (VnV)** - The process of ensuring that a design satisfies its requirements, and its requirements are aligned with the design's purpose (generally).
- **Feature-Oriented Domain Analysis (FODA)** - A domain analysis method, focusing on features of a design and their variations.
- **Proof of Concept (POC)** - An initial version of a design, intended to demonstrate the possibility of its implementation without being a full prototype.

E.2 Components

Included below is a System Context Diagram, outlining at a high level how the system will interact with external components:

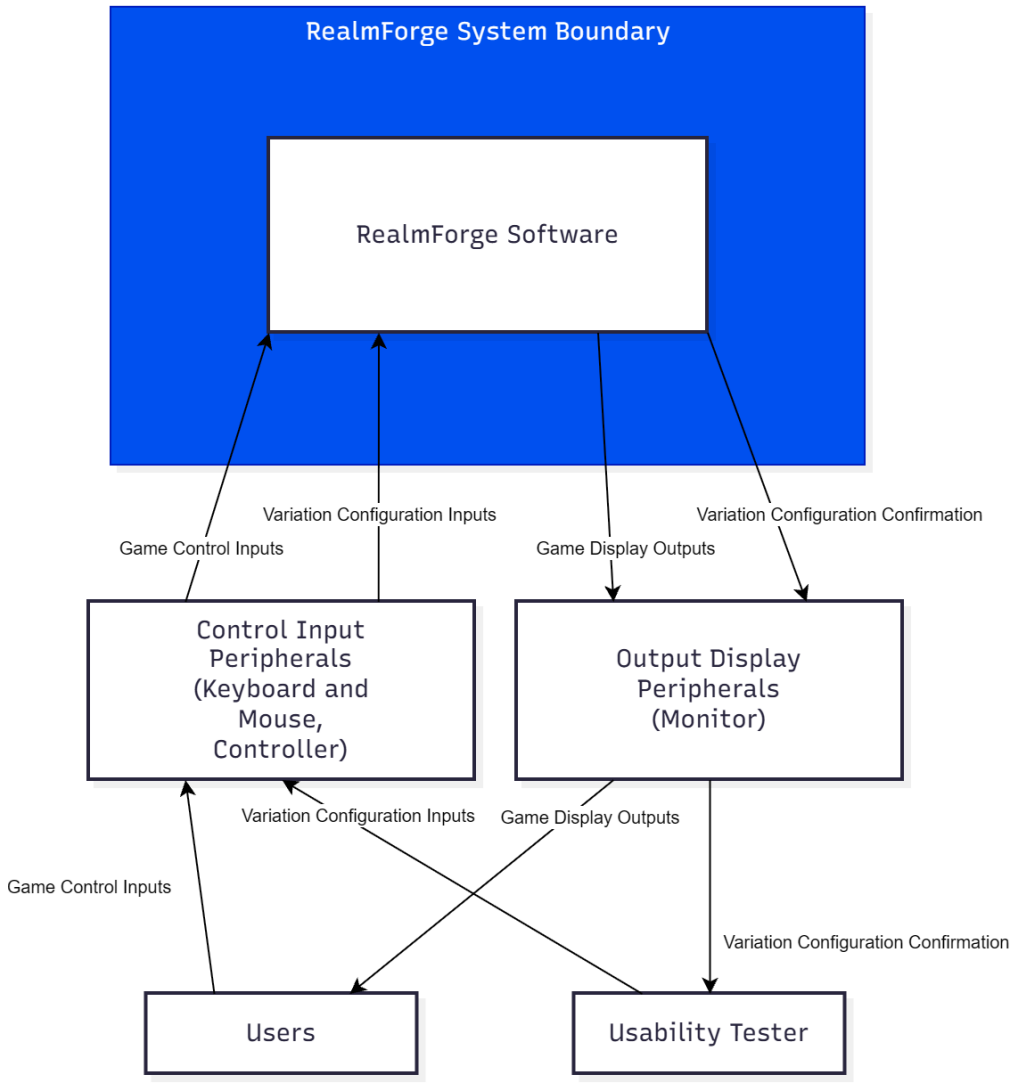


Figure 1: RealmForge System Context Diagram

The RealmForge software (also denoted as the system) will require the use of the following external components to operate successfully:

- **Control Input Peripherals:** Devices used to provide input to the system. This could include a keyboard and mouse, or a controller used to provide input. The system will require inputs from these devices to affect the state of the video game it

executes (during gameplay), and to configure the system's feature variation points during testing.

- **Output Display Peripherals:** A device used to display outputs provided by the system. This could be fulfilled by a monitor, or another display device (such as a television). The system will require this device to display graphical outputs during gameplay (video game graphics), as well as confirmation of feature variation configurations (variation config. menu graphics).

E.3 Constraints

ENV-001: The system shall be compatible with user machines running Windows 10 or 11 (64-bit) operating systems that meet the following recommended specifications:

- **CPU:** x86_64 CPU with SSE4.2 support, with 4 physical cores or more. Example: Intel Core i5-6600K, AMD Ryzen 5 1600.
- **GPU:** Dedicated graphics GPU with full OpenGL 4.6 support. Example: NVIDIA GeForce GTX 650, AMD Radeon HD 7750 (GCN 1.0).
- **RAM:** A minimum of 8 GB of RAM.

Rationale: This environment constraint is imposed to focus development of RealmForge on the most commonly used operating system for PC video games, Windows. The system may be extended to support other operating systems in the future if possible. The recommended specifications are included to ensure user machines can fully execute the system, accompanying any future decisions to extend its capabilities.

Fit Criterion: The system can be compiled and executed on machines running Windows 10 or 11 (64-bit) operating systems, meeting the recommended specifications outlined above, at a stable 60 fps.

Traceability: N/A

ENV-002: The system shall provide user input capabilities for keyboard and mouse peripherals, as well as common controllers (PlayStation, Xbox).

Rationale: The system must accommodate the use of various input methods (keyboard and mouse, different controllers) to facilitate usability studies incorporating these varying input methods.

Fit Criterion: The system can receive all game inputs required for its successful operation from the following input methods: USB Keyboard and Mouse, Xbox Series X/S, Xbox One, or Xbox 360 controllers, and PlayStation DualSense or DualShock 4 controllers.

Traceability: N/A

ENV-003: The system shall provide at least one explicit feature variation capability, that can be selected without recompiling the system's executable file.

Rationale: This constraint is imposed by the purpose of the development project (outlined in the Problem Statement document). The system must provide explicit feature variation capabilities to facilitate usability testing that will focus on the effectiveness of different variations.

Fit Criterion: The system provides at least one explicit feature variation capability, that can be selected and managed at run time, while the system is executing.

Traceability: N/A

E.4 Assumptions

Comment: *Properties of the environment that may be assumed, with the goal of facilitating the project and simplifying the system.*

E.5 Effects

Comment: *Elements and properties of the environment that the system will affect.*

E.6 Invariants

Comment: *Properties of the environment that the systemâ€™s operation must preserve.*

System

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Comment: *The System book refines the Goal one by focusing on more detailed requirements about the system under development, mainly its constituents, behaviors and properties.*

S.1 Components

Comment: *Overall structure expressed by the list of major software and, if applicable, hardware parts.*
Comment: *This chapter should not be empty!*

S.2 Functionality

Comment: *One section, S.2.n, for each of the components identified in S.2, describing the corresponding behaviors (functional and non-functional properties).* **Comment:** *This chapter should not be empty!*

S.3 Interfaces

Comment: *How the system makes the functionality of S.2 available to the rest of the world, particularly user interfaces and program interfaces (APIs).*

S.4 Detailed usage scenarios

Comment: *Examples of interaction between the environment (or human users) and the system: use cases, user stories.*

S.5 Prioritization

Comment: *Classification of the behaviors, interfaces and scenarios (S.2, S.3 and S.4) by their degree of criticality.*

S.6 Verification and acceptance criteria

Comment: *Specification of the conditions under which an implementation will be deemed satisfactory.*

Project

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Comment: *The Project book describes all the constraints and expectations not about the system itself, but about how to develop and produce it.*

P.1 Roles and personnel

Comment: *Main responsibilities in the project; required project staff and their needed qualifications.*

P.2 Imposed technical choices

Comment: *Any a priori choices binding the project to specific tools, hardware, languages or other technical parameters.*

P.3 Schedule and milestones

The project's primary schedule will follow the milestones and dates below, each named according to the deliverables they must produce (as discussed in the project's development plan):

- M1 Problem Statement, Development Plan, PoC Plan, and Team Charter (Due by September 28, 2026)
- M2 SRS and Hazard Analysis Revision 0 (Due by October 19, 2026)
- M3 VnV Plan Revision 0 (Due by November 2, 2026)
- M4 Design Document Revision -1 and CD-Fall (FODA) (Due by November 16, 2026)

- M5 PoC Demo (Due by November 23, 2026)
- M6 Design Documentation Revision 0 (Due by January 18, 2027)
- M7 Revision 0 Demo (Due by February 1, 2027)
- M8 VnV Report Revision 0 and CD-Winter Revision 0 (Usability Report) (Due by March 8, 2027)
- M9 Final Demonstration Revision 1, EXPO Demonstration (Due by March 15, 2027)
- M10 Final Documentation (Revision 1) (Due by March 31, 2027)

P.4 Tasks and deliverables

The tasks required to be completed to develop RealmForge (and its feature variation capabilities) will be managed using an iterative and incremental approach. Within each milestone (in this context the milestones associated with source code development), team members will divide the work to be done into suitable issues, viewable in RealmForge's GitHub Project. The issues will be coordinated using a Kanban method, tracking their progress and review status. The software features associated with these issues will be regularly planned, developed, reviewed, tested, and deployed using a feature branch workflow, contributing towards each set milestone.

A schedule of development deliverables pertaining to the milestones listed above is available below, complete with relevant requirement sets that are planned to be implemented at the completion of each deliverable. Milestones that purely involve documentation development are omitted, as this schedule is intended to outline the completion of software development tasks and their associated requirements.

Detailed Schedule:

- M5 PoC Demo
  - **Deadline:** November 23, 2026
  - **Software Deliverables:** RealmForge Proof of Concept Build
    - **Associated Software Tasks:**
    - Review PoC plan and requirements
    - Develop PoC features
    - Review implemented PoC features on a recurring basis during development
    - Test PoC features
    - Verify associated requirements (outlined below)

    - Deploy PoC
  - **High Priority Requirements:** NEED TO ADD
  - **Requirement Sets Implemented:** NEED TO ADD

- M7 Revision 0 Demo
  - **Deadline:** February 1, 2027
  - **Software Deliverables:** RealmForge Revision 0 Build
    - **Associated Software Tasks:**
    - Review Revision 0 design documentation and requirements
    - Develop Revision 0 features
    - Review implemented Revision 0 features on a recurring basis during development
    - Test Revision 0 features
    - Verify associated requirements (outlined below)
    - Deploy Revision 0
  - **High Priority Requirements:** NEED TO ADD
  - **Requirement Sets Implemented:** NEED TO ADD

- M9 Final Demonstration Revision 1, EXPO Demonstration
  - **Deadline:** March 15, 2027
  - **Software Deliverables:** RealmForge Revision 1 Build
    - **Associated Software Tasks:**
    - Review Revision 1 design documentation (in progress) and requirements
    - Develop Revision 1 features
    - Review implemented Revision 1 features on a recurring basis during development
    - Test Revision 1 features
    - Verify associated requirements (outlined below)
    - Deploy Revision 1
  - **High Priority Requirements:** NEED TO ADD
  - **Requirement Sets Implemented:** NEED TO ADD

P.5 Required technology elements

Comment: *External systems, hardware and software, expected to be necessary for building the system.*

P.6 Risks and mitigation analysis

Comment: *Potential obstacles to meeting the schedule of P.4, and measures for adapting the plan if they do arise.*

P.7 Requirements process and report

Comment: *Initially, description of what the requirements process will be; later, report on its steps.*

Appendix

Appendix — Generative AI Use Disclosure

AI Tools Used

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How and Where Used

Verification of AI-Generated Content

Appendix — Reflection

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How many of your requirements were inspired by speaking to your client(s) or their proxies (e.g. your peers, stakeholders, potential users)?
4. Which of the courses you have taken, or are currently taking, will help your team to be successful with your capstone project.
5. What knowledge and skills will the team collectively need to acquire to successfully complete this capstone project? Examples of possible knowledge to acquire include domain specific knowledge from the domain of your application, or software engineering knowledge, mechatronics knowledge or computer science knowledge. Skills may be related to technology, or writing, or presentation, or team management, etc. You should look to identify at least one item for each team member.
6. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?